SWYNEX – Biological Dataset Selection

1. Introduction

Bioinformatics combines biology and computational techniques to study biological data. Public genomic databases provide valuable information for biological research.

2. Dataset Information

- Dataset Size: One complete reference genome containing 29,903 nucleotides.
- Dataset Name: SARS-CoV-2 Reference Genome
- Database: NCBI GenBank
- Organism: Severe acute respiratory syndrome coronavirus 2
- Accession Number: NC_045512.2
- Sequence Length: 29,903 nucleotides
- Molecule Type: Single-stranded RNA (ssRNA)

3. Data Source

The dataset was obtained from the NCBI Nucleotide Database.

Dataset URL: https://www.ncbi.nlm.nih.gov/nuccore/NC_045512.2

4. Research Question

How can the SARS-CoV-2 genomic sequence be studied to identify its genes and understand its genomic organization?

5. Biological Significance

The SARS-CoV-2 reference genome contains genetic information required for viral replication and protein production. Studying this genome helps researchers understand viral genes and genomic organization.

6. Expected Outcome

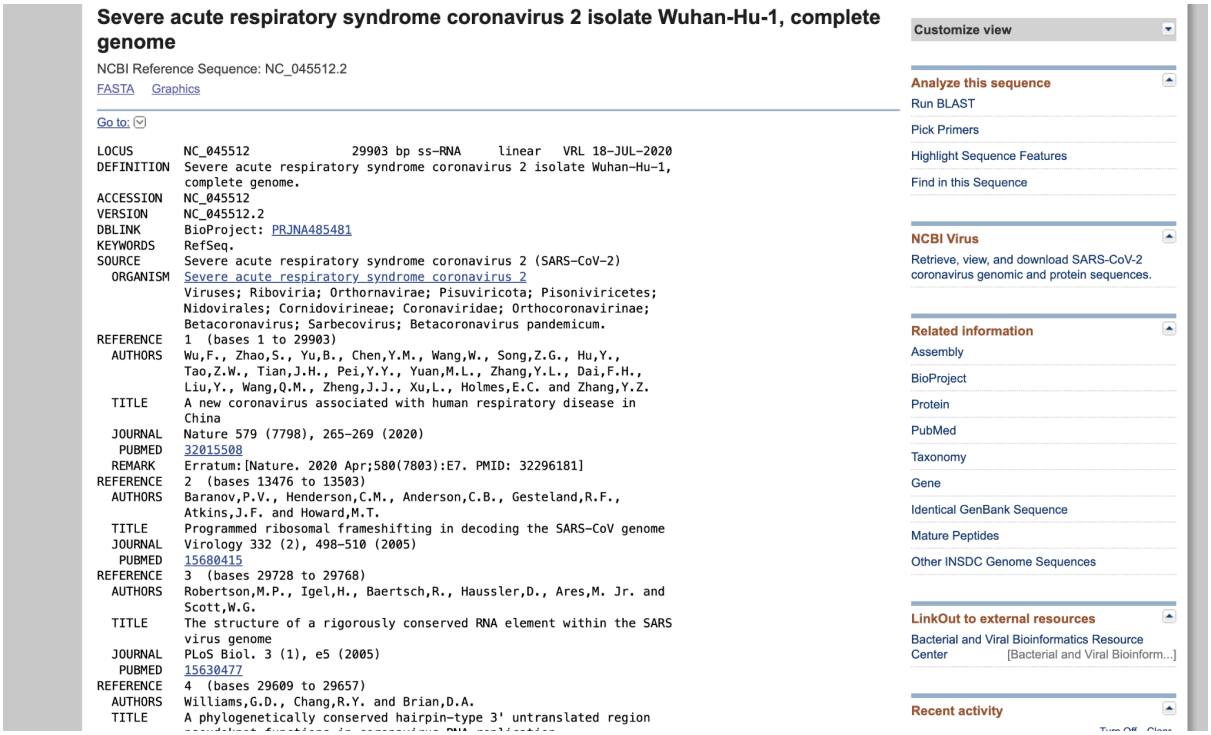
This dataset can be used to explore genomic information, identify sequence features, and learn basic bioinformatics analysis techniques.

7. Conclusion

The SARS-CoV-2 reference genome is a publicly accessible genomic resource suitable for bioinformatics learning and further computational analysis.

Reference

NCBI Nucleotide Database: NC_045512.2



Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1, complete genome

NCBI Reference Sequence: NC_045512.2

FASTA Graphics

Go to:

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LOCUS       NC_045512              29903 bp ss-RNA     linear   VRL 18-JUL-2020
DEFINITION  Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1,
            complete genome.
ACCESSION   NC_045512
VERSION     NC_045512.2
DBLINK      BioProject: PRJNA485481
KEYWORDS    RefSeq.
SOURCE      Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)
  ORGANISM  Severe acute respiratory syndrome coronavirus 2
            Viruses; Riboviria; Orthornavirae; Pisuviricota; Pisoniviricetes;
            Nidovirales; Cornidovirineae; Coronaviridae; Orthocoronavirinae;
            Betacoronavirus; Sarbecovirus; Betacoronavirus pandemicum.
REFERENCE   1  (bases 1 to 29903)
  AUTHORS   Wu,F., Zhao,S., Yu,B., Chen,Y.M., Wang,W., Song,Z.G., Hu,Y.,
            Tao,Z.W., Tian,J.H., Pei,Y.Y., Yuan,M.L., Zhang,Y.L., Dai,F.H.,
            Liu,Y., Wang,Q.M., Zheng,J.J., Xu,L., Holmes,E.C. and Zhang,Y.Z.
  TITLE     A new coronavirus associated with human respiratory disease in
            China
  JOURNAL   Nature 579 (7798), 265-269 (2020)
   PUBMED   32015508
  REMARK    Erratum:[Nature. 2020 Apr;580(7803):E7. PMID: 32296181]
REFERENCE   2  (bases 13476 to 13503)
  AUTHORS   Baranov,P.V., Henderson,C.M., Anderson,C.B., Gesteland,R.F.,
            Atkins,J.F. and Howard,M.T.
  TITLE     Programmed ribosomal frameshifting in decoding the SARS-CoV genome
  JOURNAL   Virology 332 (2), 498-510 (2005)
   PUBMED   15680415
REFERENCE   3  (bases 29728 to 29768)
  AUTHORS   Robertson,M.P., Igel,H., Baertsch,R., Haussler,D., Ares,M. Jr. and
            Scott,W.G.
  TITLE     The structure of a rigorously conserved RNA element within the SARS
            virus genome
  JOURNAL   PLoS Biol. 3 (1), e5 (2005)
   PUBMED   15630477
REFERENCE   4  (bases 29609 to 29657)
  AUTHORS   Williams,G.D., Chang,R.Y. and Brian,D.A.
  TITLE     A phylogenetically conserved hairpin-type 3' untranslated region
```

Customize view

Analyze this sequence
Run BLAST
Pick Primers
Highlight Sequence Features
Find in this Sequence

NCBI Virus
Retrieve, view, and download SARS-CoV-2 coronavirus genomic and protein sequences.

Related information
Assembly
BioProject
Protein
PubMed
Taxonomy
Gene
Identical GenBank Sequence
Mature Peptides
Other INSDC Genome Sequences

LinkOut to external resources
Bacterial and Viral Bioinformatics Resource Center [Bacterial and Viral Bioinform...]

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